NAME: Rohan Garg

ROLLNO:1024150187

ASSIGNMENT 3

Q1. Write a program to implement (a) pointer to an object (b) this pointer. Practice both '.' (dot operator) and '->' (arrow operator).

```
#include <iostream>
using namespace std;

class Student {
public:
    int roll;
    string name;

    void display() {
        cout << "Roll: " << roll << endl;
        cout << "Name: " << name << endl;
    }
};

int main() {
    Student s1;
    s1.roll = 219;
    s1.name = "Sara";

    Student *ptr;
    ptr = &s1;

    cout << "Using dot operator:" << endl;
    s1.display();

    cout << "\nUsing arrow operator:" << endl;
    ptr->display();

    return 0;
}
```

```
Using dot operator:
Roll: 219
Name: Sara

Using arrow operator:
Roll: 219
Name: Sara
```

Q2. Write a program to swap private values of two classes using a friend function.

```cpp
#include <iostream>
using namespace std;

class B;

class A {
    int x;
public:
    void set(int a) {
        x = a;
    }
    void show() {
        cout << "Value of x in class A: " << x << endl;
    }
    friend void swap(A &, B &);
};

class B {
    int y;
public:
    void set(int b) {
        y = b;
    }
    void show() {
        cout << "Value of y in class B: " << y << endl;
    }
    friend void swap(A &, B &);
};
```

```cpp
void swap(A &obj1, B &obj2) {
    int temp;
    temp = obj1.x;
    obj1.x = obj2.y;
    obj2.y = temp;
}

int main() {
    A obj1;
    B obj2;

    obj1.set(10);
    obj2.set(20);

    cout << "Before swapping:" << endl;
    obj1.show();
    obj2.show();

    swap(obj1, obj2);

    cout << "\nAfter swapping:" << endl;
    obj1.show();
    obj2.show();

    return 0;
}
```

```
Before swapping:
Value of x in class A: 10
Value of y in class B: 20

After swapping:
Value of x in class A: 20
Value of y in class B: 10
```

Q3.Write a program to add data objects of two different classes using friend functions.

```cpp
#include <iostream>
using namespace std;

class B;

class A {
    int x;
public:
    void set(int a) {
        x = a;
    }
    friend void add(A, B);
};

class B {
    int y;
public:
    void set(int b) {
        y = b;
    }
    friend void add(A, B);
};
void add(A obj1, B obj2) {
    int sum;
    sum = obj1.x + obj2.y;
    cout << "Sum = " << sum << endl;
}
int main() {
    A obj1;
    B obj2;

    obj1.set(10);
    obj2.set(20);

    add(obj1, obj2);
    return 0;
}
```

```
PS C:\Users\sarag\OneDrive\Desktop\c++> g++ sort.cpp
PS C:\Users\sarag\OneDrive\Desktop\c++> ./a.exe
Sum = 30
```

Q4.Write a program to demonstrate the working of friend class.

```cpp
#include <iostream>
using namespace std;

class B;

class A {
    int x;
public:
    void set(int a) {
        x = a;
    }
    friend class B;
};
class B {
public:
    void show(A obj) {
        cout << "Value of x in class A: " << obj.x << endl;
    }
};
int main() {
    A objA;
    B objB;

    objA.set(50);
    objB.show(objA);
    return 0;
}
```

```
PS C:\Users\sarag\OneDrive\Desktop\c++> g++ sort.cpp
PS C:\Users\sarag\OneDrive\Desktop\c++> ./a.exe
Value of x in class A: 50
```

Q5.Write a program using Array of Objects to display area of multiple rectangles.

```cpp
#include <iostream>
using namespace std;

class Rectangle {
    int length, breadth;
public:
    void set(int l, int b) {
        length = l;
        breadth = b;
    }

    void area() {
        cout << "Area = " << length * breadth << endl;
    }
};

int main() {
    int n;
    cout << "Enter number of rectangles: ";
    cin >> n;

    Rectangle r[n];

    for(int i = 0; i < n; i++) {
        int l, b;
        cout << "Enter length and breadth of rectangle " << i+1 << ": ";
        cin >> l >> b;
        r[i].set(l, b);
    }

    cout << "\nAreas of rectangles:\n";
    for(int i = 0; i < n; i++) {
        r[i].area();
    }
    return 0;
}
```

```
Enter number of rectangles: 4
Enter length and breadth of rectangle 1: 6
8
Enter length and breadth of rectangle 2: 4 5
Enter length and breadth of rectangle 3: 7 2
Enter length and breadth of rectangle 4: 9 3

Areas of rectangles:
Area = 48
Area = 20
Area = 14
Area = 27
```

Q6.Write a program to define function cube() as inline for calculating cube of a number.

```cpp
#include <iostream>
using namespace std;

inline int cube(int x) {
    return x * x * x;
}

int main() {
    int num;
    cout << "Enter a number: ";
    cin >> num;

    cout << "Cube of " << num << " is: " << cube(num) << endl;

    return 0;
}
```

```
PS C:\Users\sarag\OneDrive\Desktop\c++> g++ sort.cpp
PS C:\Users\sarag\OneDrive\Desktop\c++> ./a.exe
Enter a number: 3
Cube of 3 is: 27
```

Q7.Write a program to pass an object as an argument and return the object from a function**.**

a. Use pass-by-value

b. Use pass-by-address

```cpp
#include <iostream>
using namespace std;

class Number {
public:
    int x;

    void set(int a) {
        x = a;
    }

    void show() {
        cout << "Value: " << x << endl;
    }
};

Number increment(Number obj) {
    obj.x = obj.x + 10;
    return obj;
}
int main() {
    Number n1, n2;
    n1.set(5);

    n2 = increment(n1);

    cout << "Original object: ";
    n1.show();

    cout << "Returned object: ";
    n2.show();
    return 0;
}
```

```
Original object: Value: 5
Returned object: Value: 15
```

b)

```cpp
#include <iostream>
using namespace std;

class Number {
public:
    int x;

    void set(int a) {
        x = a;
    }

    void show() {
        cout << "Value: " << x << endl;
    }
};

Number increment(Number *obj) {
    obj->x = obj->x + 10;
    return *obj;
}

int main() {
    Number n1, n2;
    n1.set(5);

    n2 = increment(&n1);

    cout << "Original object: ";
    n1.show();

    cout << "Returned object: ";
    n2.show();

    return 0;
}
```

```
PS C:\Users\sarag\OneDrive\Desktop\c++> g++ sort.cpp
PS C:\Users\sarag\OneDrive\Desktop\c++> ./a.exe
Original object: Value: 15
Returned object: Value: 15
```